
CprE 458/558: Real-Time Systems

Real-Time Networks – WAN
Traffic Shaping/Policing

Regulating flow control

- The bursty traffic in the network results in congestion
- Traffic shaping reduces congestion and thus helps the carrier live up to its guarantees
- Traffic shaping is about regulating the average rate (and burstiness) of data transmission

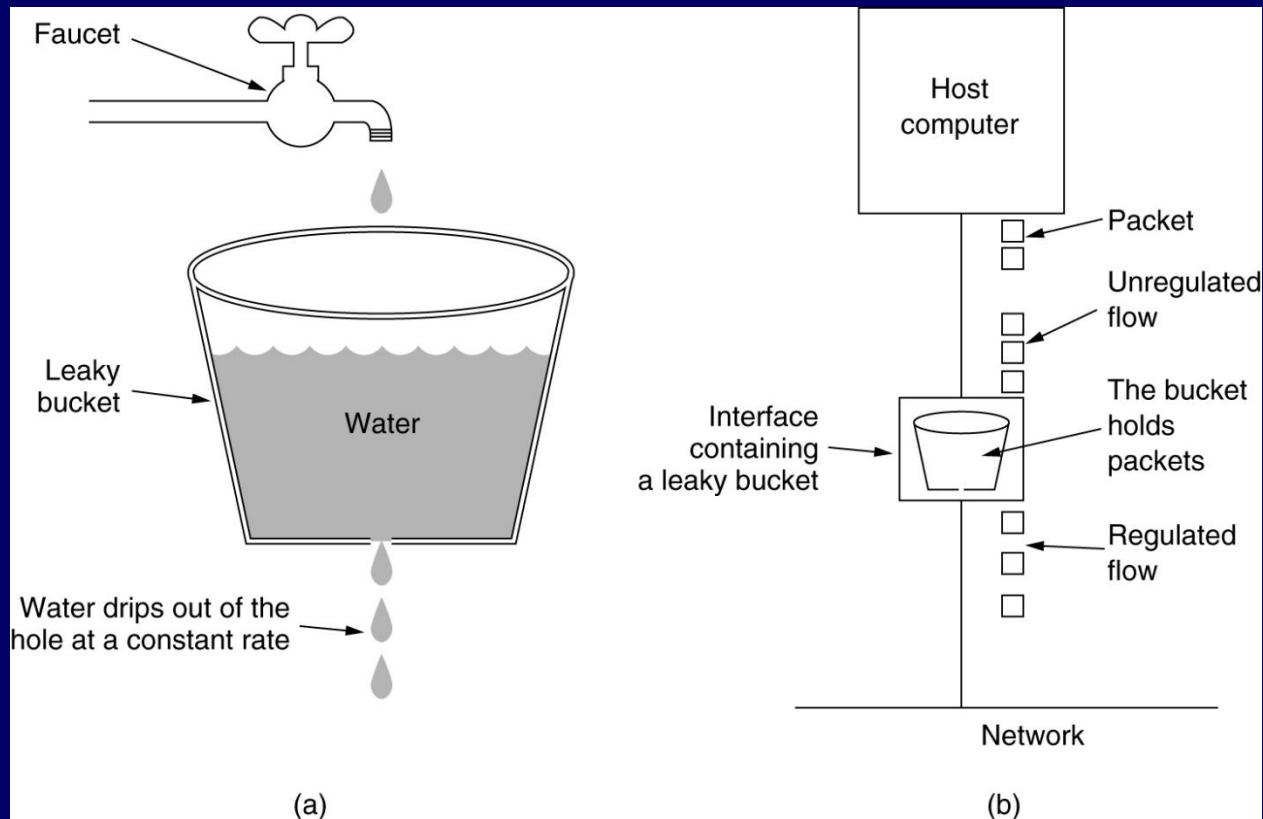
Traffic Shaping

- Traffic shaping controls the *rate* at which packets are sent (not just how many)
- At connection set-up time, the sender and carrier negotiate a traffic pattern (shape)
- Two traffic shaping algorithms are:
 - Leaky Bucket
 - Token Bucket

The Leaky Bucket Algorithm

- The **Leaky Bucket Algorithm** used to control rate in a network. It is implemented as a single-server queue with constant service time. If the bucket (buffer) overflows then packets are discarded.

The Leaky Bucket Algorithm



(a) A leaky bucket with water. (b) a leaky bucket with packets.

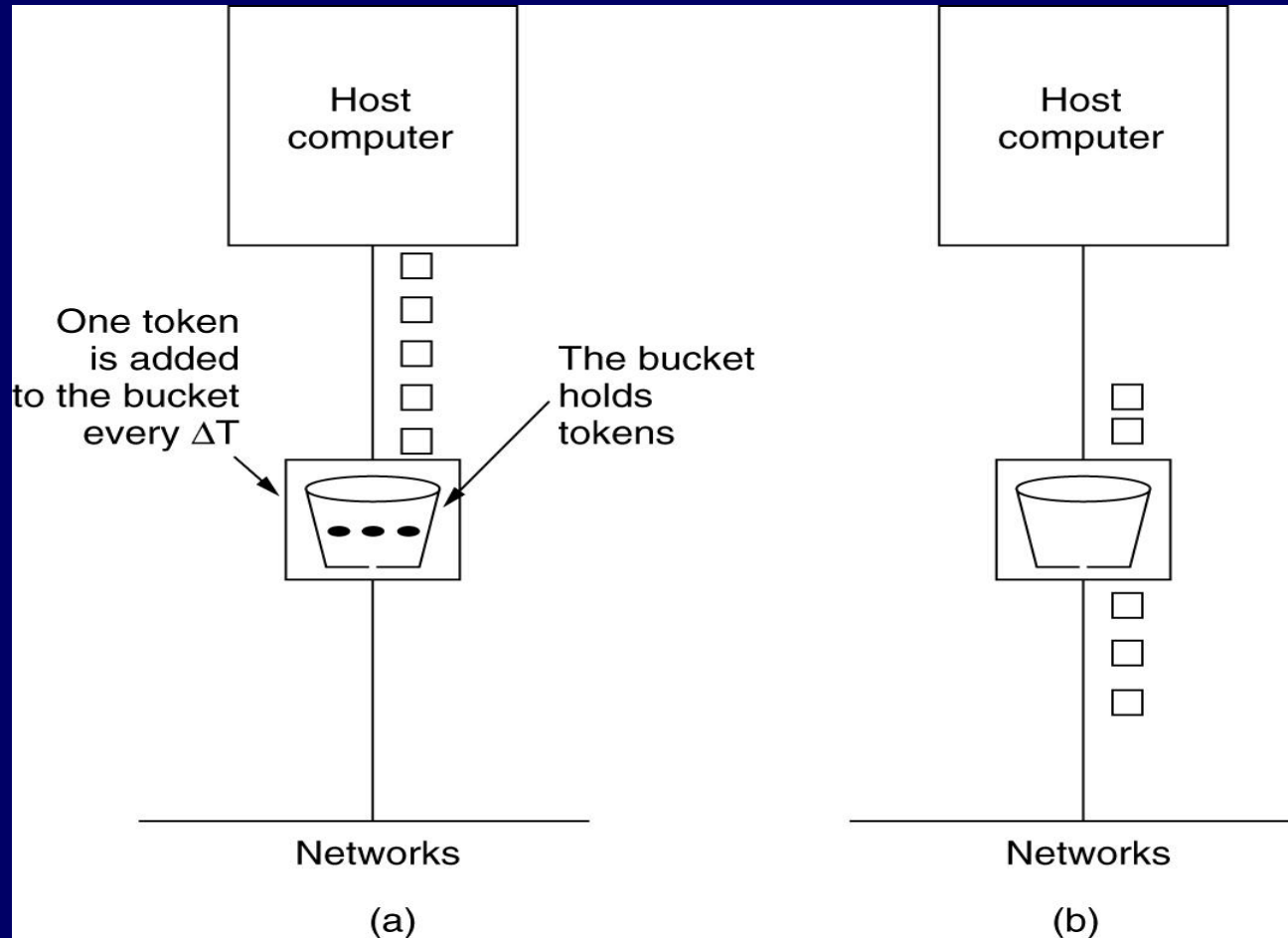
Leaky Bucket Algorithm (contd.)

- The leaky bucket enforces a constant output rate regardless of the burstiness of the input. Does nothing when input is idle.
- The host injects one packet per clock tick onto the network. This results in a uniform flow of packets, smoothing out bursts and reducing congestion.
- When packets are the same size (as in ATM cells), the one packet per tick is okay. For variable length packets though, it is better to allow a fixed number of bytes per tick.

Token Bucket Algorithm

- In contrast to the LB, the Token Bucket (TB) algorithm, allows the output rate to vary, depending on the size of the burst.
- In the TB algorithm, the bucket holds tokens. To transmit a packet, the host must capture and destroy one token.
- Tokens are generated by a clock at the rate of one token every Δt sec.
- Idle hosts can capture and save up tokens (up to the max. size of the bucket) in order to send larger bursts later.

Token Bucket Algorithm (contd.)



(a) Before (b) After

Token bucket operation

- TB accumulates fixed size tokens in a token bucket
- Transmits a packet (from data buffer, if any are there) or arriving packet if the sum of the token sizes in the bucket add up to packet size
- More tokens are periodically added to the bucket (at rate Δt). If tokens are to be added when the bucket is full, they are discarded

Token bucket properties

- Does not bound the peak rate of small bursts, because bucket may contain enough token to cover a complete burst size
- Performance depends only on the sum of the data buffer size and the token bucket size

Token bucket - example

- 2 tokens of size 100 bytes added each second to the token bucket of capacity 500 bytes
 - Avg. rate = 200 bytes/sec, burst size = 500 bytes
 - Packets bigger than 500 bytes will never be sent
 - Peak rate is unbounded – i.e., 500 bytes of burst can be transmitted arbitrarily fast

Leaky Bucket vs Token Bucket

- LB discards packets; TB does not. TB discards tokens.
- With TB, a packet can only be transmitted if there are enough tokens to cover its length in bytes.
- LB sends packets at an average rate. TB allows for large bursts to be sent faster by speeding up the output.
- TB allows saving up tokens (permissions) to send large bursts. LB does not allow saving.